

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

November 2013

EE305 Power Electronics & Drives I

Date: 27.11.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Draw and explain static I-V characteristics of a Thyristor. [07]

Q - 2 (a) The specification sheet for an SCR gives maximum rms on-state current as 15 A. If this SCR is used in a Resistive circuit, compute average on-state current rating for half-sine wave current for conduction angles of 180° and 30° . [07]

(b) Draw and explain I-V characteristics of Triac in mode I⁺. [07]

OR

Q - 2 (a) Draw and explain Switching characteristics of IGBT. [07]

(b) Explain three-phase full-wave controlled rectifier with R load with all necessary waveforms and circuit diagram. [07]

Q - 3 (a) Draw the only circuit and required waveforms for Complementary Commutation of a Thyristor. [07]

(b) Compare GTO and SCR. [04]

(c) Draw cross-sectional view, circuit symbol and I-V characteristics of DIAC. [03]

OR

Q - 3 (a) Draw and explain Second-quadrant or Type-B Chopper. [06]

(b) Draw and explain Single-phase Dual converter drives for separately-excited DC motor. [08]

SECTION – II

- Q - 4 A Relaxation Oscillator using UJT, as shown in Fig-1 is to be designed for triggering of an SCR. The UJT has following data: [07]

$\eta = 0.72$, $I_p = 0.6 \text{ mA}$, $V_p = 18.0 \text{ V}$, $V_v = 1.0 \text{ V}$, $I_v = 2.5 \text{ mA}$, $R_{BB} = 5 \text{ k}\Omega$, normal leakage current with emitter open = 4.2 mA .

The firing frequency is 2 kHz . For $C = 0.04 \mu\text{F}$, compute the values of R , R_1 and R_2 .

- Q - 5 (a) Explain Single-phase half-wave controlled rectifier circuit with R-L load. Draw neat circuit diagram and waveforms. Also find out the equation of rms output voltage average output voltage. [10]
- (b) Write down the Advantages and disadvantages of Power Electronic Converters. [04]

OR

- Q - 5 (a) A Step-up chopper has input voltage of 220 V and output voltage of 660 V . If the conducting time of the chopper is $100 \mu\text{s}$, compute the pulse width of output voltage. In case output voltage pulse width is halved for constant frequency operation, find the average value of new output voltage. [05]
- (b) Explain concept of Modern Electric drive using Power Electronic Converters. [05]
- (c) Explain Time ratio Control (TRC) for varying duty cycle in chopper. [04]
- Q - 6 (a) Explain the principle of operation of Chopper with neat diagram. [07]

- (b) The circuit as shown in Fig-2 is initially relaxed. The thyristor T is turned on at $t=0$. Determine (a) conduction time of thyristor and (b) voltage across thyristor and capacitor after SCR is turned off. Calculate these values for $L = 5 \text{ mH}$, $C = 20 \mu\text{F}$ and $V_s = 200 \text{ V DC}$. Draw all necessary waveforms. [07]

OR

- Q - 6 (a) A single-phase transformer with secondary voltage of 230 V , 50 Hz , delivers power to load $R = 20 \Omega$ through a half-wave controlled rectifier circuit. For a firing-angle delay of 60° , determine (i) output average voltage and current (ii) output rms voltage and current (iii) Rectification efficiency. [07]
- (b) Explain single-phase half-wave converter drive for separately-excited DC motor. [07]

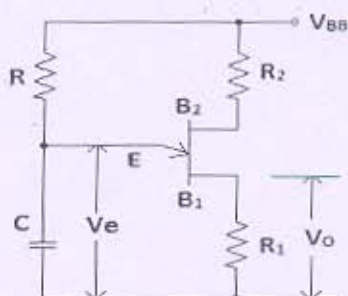


Fig-1

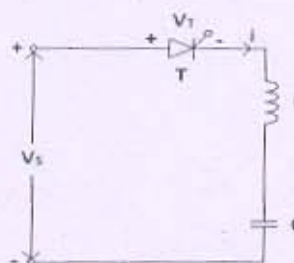


Fig-2
